

C-CSF: Accurate, robust and efficient surface tension and contact angle models for single-phase flows using SPH

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Abstract

In this paper we propose some important improvements related to the single-phase Smoothed Particle Hydrodynamics (SPH) simulations and available for different SPH schemes (such as Riemann SPH or δ -SPH). This model is based on the Continuous Surface Force (CSF), a volumic description of the surface tension and relies on an accurate evaluation of the local normal and curvature at the interface. In particular we show that renormalized SPH operators are necessary to both improve the stability and the accuracy of the model. We also propose a solution aiming at imposing a desired equilibrium contact angle near the contact line for the Boundary Integral Method (BIM). The proposed model is called corrected CSF (C-CSF) and is validated through static and dynamic test cases, with and without solid boundaries, showing that the proposed corrections are necessary to obtain the expected solutions.

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1. Introduction

Surface tension effects take place in a wide range of physical and industrial problems such as microfluidics, film flows or drops formation. While the effects of surface tension can be neglected in many cases they become significant as soon as the characteristic size of the problem is sufficiently small. For those particular cases, the surface tension must be taken into account accurately to properly simulate the considered flows.

From the physical point of view, the surface tension is a force occurring at the interface between two fluids. This force can be explained in two different ways, depending on whether a microscopic (molecular level) or a macroscopic (continuous level) description is adopted. At the molecular level, surface tension is explained by an imbalance of the interactions between the molecules of the two fluids (and/or fluid–solid) in contact. Typically the range of those imbalanced interactions is about a few times the characteristic size of the molecules. At the macroscopic level, surface tension is generally explained as a surfacic and continuous force occurring at the interface between two different fluids.

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For numerical simulations involving surface tension effects, those two points of view can be adopted, each with advantages and drawbacks. Since one generally solves the equations of the continuum mechanics, the macroscopic (continuous) description of the surface tension is usually preferred. Most of the numerical methods rely on a volumic description of the computational domain, such as Finite Volume or Smoothed Particle Hydrodynamics (SPH) methods. This volumetric description generally makes the detection of the interface between two fluids difficult to achieve so that the implementation of surface tension effects is not straightforward. Nevertheless this drawback can be bypassed by rewriting the classical surfacic formulation of the surface tension into a volumic description which can be implemented more easily into volumic numerical methods. This reformulation has been first proposed by Brackbill et al. [1] and is based on the computation of the curvature at the interface, combined with the use of a color gradient. This formulation is often call the Continuous Surface Force (CSF) model [1]. It has been then used in several numerical methods [1–4] [5]. As showed in [1], this formulation converges towards the surfacic and continuous description of the surface tension as the discretization size decreases. In SPH, this approach was first tested by Morris [6] and then extended by Hu and Adams [7]. This approach has been validated for multiphase flows, but given what we know about this approach has not been validated in single-phase simulation for SPH method. These two approaches can also take into account the phenomenon of contact angle at the contact line by imposing it at the contact line [7,8].

On the other hand the “molecular” point of view of the phenomenon can also be adopted in numerical simulation. As explained above, in molecular scale the surface tension is caused by an imbalance of the interactions between molecules at the interface. Therefore, a way to impose surface tension effects is to mimic this imbalance by imposing a specific Pairwise Force (PF) field near the interface between the two fluids. In the SPH method, its particle-based feature eases the implementation of the “molecular” approach described above (see [9–11] for more details). This approach is easy to implement but it generally requires a calibration phase which could depend on the considered problem. However Tartakovsky and Panchenko [12] recently derived a relationship between the pairwise force field and the surface tension coefficients, avoiding the calibration phase. as the main advantage of this approach, it is totally valid for single-phase simulation, strongly decreasing the computational costs. But this approach provides some non-physical pressures at the interface due to the inconsistency between the SPH particles and the molecular spacing scale at which the surface tension occurs. For this reason, we did not retained the PF-SPH model in the present work.

This paper aims to propose and validate a corrected CSF (C-CSF) formulation of the surface tension available in single-phase flows and for various SPH schemes (such as Riemann SPH [13] or δ -SPH [14,15]). We also proposed a model in the presence of solid boundaries and in particular for the contact angle phenomenon. We mainly develop the model of the contact angle with the Cutface approach (BIM-CFA) [16], which belongs to the class of Boundary Integral Method (BIM) [17,18], because it is more suitable for cases involving complex geometries. The ghost particle formulation is also explored.

This paper is structured as follows. We first detail the C–CSF formulation for simulations without solid boundaries and validate it with various test cases. Then modifications are proposed to include solid boundaries while incorporating the contact angle modeling. This approach is then discussed and validated.

2. Governing equations

As shown in [19], all SPH schemes can be decomposed into three different parts : a Lagrangian part, a stabilization part and a shifting part. Then a Lagrangian SPH scheme is expressed as :

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{u}_i + 0 + \delta\mathbf{u}_i, \quad (1)$$

$$\frac{dV_i}{dt} = V_i \sum_{j \in \Omega_i} (\mathbf{u}_j - \mathbf{u}_i) \cdot \nabla W_{ij} V_j + \Pi_i^{(V)} + 0, \quad (2)$$

$$\frac{d(V_i \rho_i)}{dt} = 0 + 0 + 0, \quad (3)$$

$$\frac{d(V_i \rho_i \mathbf{u}_i)}{dt} = \underbrace{-V_i \sum_{j \in \Omega_i} (P_j + P_i) \nabla W_{ij} V_j + V_i \rho_i \mathbf{g} + V_i \mathbf{F}_i^\mu + V_i \mathbf{F}_i^\sigma}_{\text{Lagrangian part}} + \underbrace{\Pi_i^{(\rho V \mathbf{u})}}_{\text{stabilization part}} + \underbrace{\mathbf{0}}_{\text{shifting part}}, \quad (4)$$

where $\mathbf{x}_i$, $\mathbf{u}_i$, $\delta\mathbf{u}_i$, V_i , ρ_i and P_i are respectively the position, the Lagrangian velocity, the shifting velocity, the discrete volume, the density and the pressure of the particle i . W is the SPH kernel with a compact support of radius R . In this work the Wendland- C^2 kernel is used [20,21]. $\mathbf{F}_i^\mu$ and $\mathbf{F}_i^\sigma$ are respectively the viscous and surface tension forces. Note that this latter force is written here using a volumic description. The stabilization terms $\Pi_i^{(V)}$ and $\Pi_i^{(\rho V \mathbf{u})}$ depend on the choice of the numerical SPH scheme. In this work the Riemann-based SPH scheme from Parshikov et al. [13] is mostly used. For this Riemann-based scheme, the stabilization terms write [19] :

$$\Pi_i^{(V)} = \Pi_{i,Rie}^{(V)} = V_i \sum_{j \in \Omega_i} (2\mathbf{u}_E - \mathbf{u}_j - \mathbf{u}_i) \cdot \nabla W_{ij} V_j, \quad (5)$$

$$\Pi_i^{(\rho V \mathbf{u})} = \Pi_{i,Rie}^{(\rho V \mathbf{u})} = -V_i \sum_{j \in \Omega_i} (2P_E - P_j - P_i) \nabla W_{ij} V_j, \quad (6)$$

where the subscript E refers to the numerical solution of the Riemann problem between particles i and j , solved at the location $\mathbf{x}_{ij} = \frac{\mathbf{x}_i + \mathbf{x}_j}{2}$. Even though an exact Riemann solver can be used, here a linearized Riemann solver [22] is adopted to determine P_E and $\mathbf{u}_E$:

$$\begin{cases} \mathbf{u}_E = \left[\frac{(\rho_L c_L \mathbf{u}_L + \rho_R c_R \mathbf{u}_R) \cdot \mathbf{n}}{\rho_L c_L + \rho_R c_R} - \frac{P_R - P_L}{\rho_L c_L + \rho_R c_R} \right] \mathbf{n} \\ P_E = \frac{\rho_L c_L P_L + \rho_R c_R P_R}{\rho_L c_L + \rho_R c_R} - \frac{\rho_L c_L \rho_R c_R (\mathbf{u}_R - \mathbf{u}_L) \cdot \mathbf{n}}{\rho_L c_L + \rho_R c_R}, \end{cases} \quad (7)$$

where $\mathbf{n}$ is the normal direction of the considered Riemann problem and ρ_L , c_L , P_L and $\mathbf{u}_L$ are respectively the density, the sound speed, the pressure and the velocity of the left state of the Riemann problem (identically subscript R refers to the right state). These left and right states are reconstructed using the MUSCL scheme [23], from particles i and j respectively. The MUSCL gradients of the conservative variables $\phi = \left(\frac{\rho}{\rho_u} \right)$ are calculated with a renormalized gradient operator as:

$$\mathbb{G}rad \phi_i^{MUSCL} = \mathbb{L}_i \left(\sum_{j \in \Omega_i} (\phi_j - \phi_i) \otimes \nabla W_{ij} V_j \right), \quad (8)$$

where the renormalization matrix $\mathbb{L}_i$ is expressed as:

$$\mathbb{L}_i = \left(\sum_{j \in \Omega_i} (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij} V_j \right)^{-1}. \quad (9)$$

The shifting velocity $\delta\mathbf{u}_i$ is used to impose a particle shifting and avoid the occurrence of Lagrangian particle structures, and this for various SPH schemes [15,24,25]. In this work the expression proposed by Oger et al. [25] is used to compute $\delta\mathbf{u}_i$.

Finally, the following equation of state is used to close the system (1)–(4):

$$P = \frac{\rho_0 c_0^2}{\gamma} \left[\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right] \quad (10)$$

where ρ_0 is the reference density, c_0 the speed of sound and $\gamma = 7$ for water. The parameter c_0 is chosen so that the density variation does not exceed 1% (or identically, so that the Mach number does not exceed 0.1) in order to maintain a weakly compressible feature (i.e. $\nabla \cdot \mathbf{u} \simeq 0$).

The viscous force $\mathbf{F}_i^\mu$, in a weakly-compressible framework, is expressed as the Laplacian of the velocity field:

$$\mathbf{F}_i^\mu = \mu \Delta \mathbf{u}_i, \quad (11)$$

where μ is the dynamic viscosity of the fluid. In the SPH method, two different Laplacian operators are commonly used to compute $\Delta \mathbf{u}_i$. The first one has been proposed by Monaghan and Gingold [26] and is expressed as:

$$\Delta^{MG} \mathbf{u}_i = C \sum_{j \in \Omega_i} \frac{(\mathbf{u}_i - \mathbf{u}_j) \cdot (\mathbf{x}_i - \mathbf{x}_j)}{\|\mathbf{x}_i - \mathbf{x}_j\|^2} \nabla W_{ij} V_j, \quad (12)$$

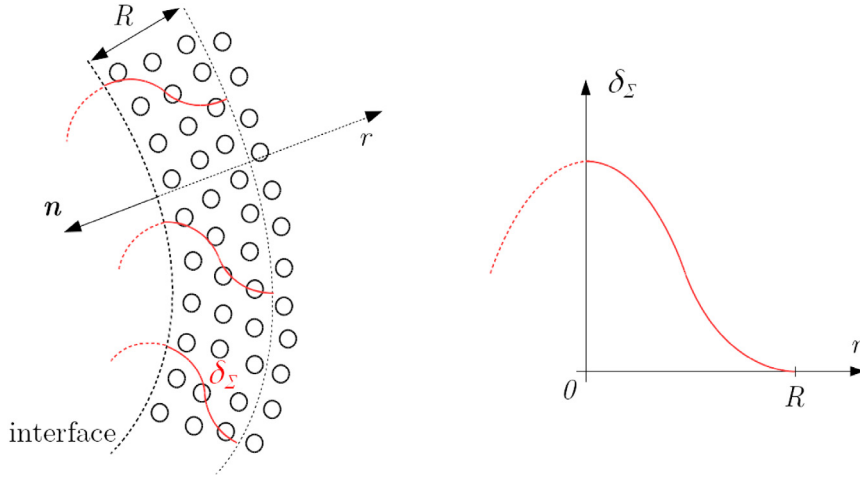


Fig. 1. Sketch of the behavior of the interfacial function ; dotted lines figure the behavior in the missing phase.

where C is a constant equal to 8 in 2D and 10 in 3D. The second one has been proposed by Morris et al. [27] and is expressed as follows:

$$\Delta^{MEA} \mathbf{u}_i = 2 \sum_{j \in \Omega_i} \frac{(\mathbf{u}_i - \mathbf{u}_j)}{\|\mathbf{x}_i - \mathbf{x}_j\|^2} (\mathbf{x}_i - \mathbf{x}_j) \cdot \nabla W_{ij} V_j. \quad (13)$$

As showed by Colagrossi et al. [28], the Laplacian operator proposed by Morris et al. is not consistent in the context of free-surface flows and therefore for flows involving surface tension effects. However, no solution has been proposed yet to combine the operator (12) with the BIM formulation for boundary conditions. In the present work, we therefore preferred the use of (13) for which Chiron et al. [16] proposed an adaptation to the BIM boundary conditions.

As proposed in [1], the surface tension force $\mathbf{F}_i^\sigma$ can be written in CSF formalism as:

$$\mathbf{F}_i^\sigma = -\sigma \kappa_i \mathbf{n}_i \delta_{\Sigma,i}, \quad (14)$$

where σ is the surface tension coefficient, κ_i the curvature of the interface, $\mathbf{n}_i$ the unit vector normal to the interface and $\delta_{\Sigma,i}$ the interfacial function. The interfacial function $\delta_{\Sigma,i}$ vanishes as the distance to the interface increases (see Fig. 1) and is around 0 when the distance to the interface is greater than a particle radius R , ensuring the consistency of this volumic formulation with the surfacic description of the surface tension [1]. Expression (14) has been built to impose the pressure jump at the interface in the context of multiphase flows (see [1] for more details). It has also been adopted in [29], in a multiphase SPH formalism. The next section of this paper is dedicated to presenting all the terms in (14) adapted to the single-phase SPH formalism.

3. Enhanced mono-fluid CSF formulation for surface tension in SPH

We propose here to adapt the CSF formulation proposed in [1] to the single-phase SPH formulation. The main difficulty encountered here is the incompleteness of the kernel support in the interface region (conversely to the multiphase case), so that the accuracy of the SPH operators in this area has a strong influence in the proposed model. The present section introduces and discusses the different terms in (14) in the single-phase context, in absence of solid boundaries. This enhanced formulation is named 'corrected CSF' (C-CSF).

3.1. Determination of the normal to the interface

We first need to evaluate the normal to the interface. The normal to the interface is usually evaluated using the color gradient at the interface [1,6,7,29]. However this evaluation remains sufficiently accurate in the immediate proximity of the interface exclusively (typically one or two times the characteristic discretization length). Ideally

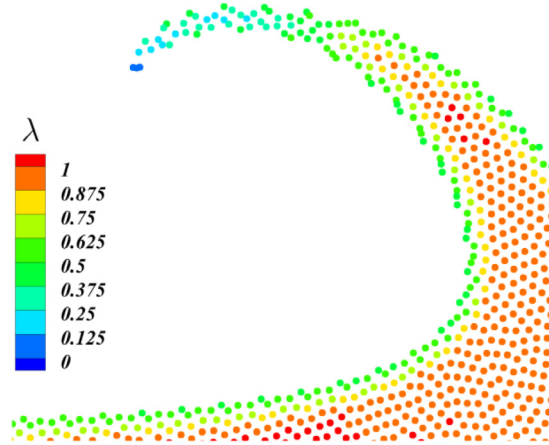


Fig. 2. Behavior of the λ field near the interface, and near a fluid jet.

the evaluation of the normal to the interface must be made up to a distance to the interface corresponding to the radius of the compact support of the kernel function. In the context of particle shifting near a free-surface, Sun et al. [15] showed that the evaluation of the normal to the interface can be improved by using the gradient of the minimum eigenvalue λ_i of the inverse renormalization matrix $\mathbb{L}_i^{-1}$.

Similarly to Huber et al. [30], we found that a renormalized gradient operator should be used to evaluate the normal to the interface, as it provides some better results (see Section 4.1). The normal to the interface is finally expressed as:

$$\mathbf{n}_i = -\frac{\nabla \lambda_i}{\|\nabla \lambda_i\|}, \quad (15)$$

where

$$\nabla \lambda_i = \sum_{j \in \Omega_i} (\lambda_j - \lambda_i) (\mathbb{L}_i \nabla W_{ij}) V_j. \quad (16)$$

The minus sign in (15) imposes the normal $\mathbf{n}_i$ to be directed outward the fluid domain. The value of λ_i is equal to 1 far from the interface (as soon as the distance from the interface exceeds the radius of the kernel) and it decreases to lower values as the distance to the interface decreases (typically $\lambda_i \simeq 0.5$ at the interface for a flat free-surface). Fig. 2 gives an overview of the behavior of the value of λ_i , especially near a fluid jet. This explains why the expression (15) provides better evaluation of the normal as the distance to the interface increases than the classical normal evaluation based on the color gradient. However it requires additional operations for the computation of λ_i and the renormalization of the gradient (16).

3.2. Curvature at the interface

Then the curvature at the interface is simply computed by determining the divergence of the previously computed normals. Here again, as discussed in [30], a renormalized operator is preferred as we find it gives some better results (see Section 4.1). The curvature κ_i is finally expressed as:

$$\kappa_i = \sum_{j \in \Omega_i} (\mathbb{L}_i (\mathbf{n}_j - \mathbf{n}_i)) \cdot \nabla W_{ij} V_j. \quad (17)$$

3.3. Interfacial function

The interfacial function $\delta_{\Sigma,i}$ is usually determined using the norm of the color gradient ∇c_i [6,29]:

$$\nabla c_i = \sum_{j \in \Omega_i} (c_j + c_i) \nabla W_{ij} V_j. \quad (18)$$

In single-phase simulations the color function is simply given by $c_i = c_j = 1$, leading to the following expression for the interfacial function:

$$\delta_{\Sigma,i} = \|\nabla c_i\| = 2 \left\| \sum_{j \in \Omega_i} \nabla W_{ij} V_j \right\|. \quad (19)$$

3.4. Critical corrections

Some corrections are necessary to make the present formulation more accurate and robust, in particular for particles having only few neighbors such as in fluid jet areas for instance (typically when the kernel support intersects one or several free-surfaces), and for which the SPH operators might provide some erroneous approximations. Note that these corrections should also preserve a null surface tension force for single particles (with no neighbors).

3.4.1. Normal correction inside fluid jets

We first notice that $\nabla \lambda_i$ is erroneous inside fluid jets and in particular that the directions of this vector are wrongly evaluated. Indeed, inside fluid jets (i.e. for low number of neighbor particles ; see Fig. 2) we observe that λ is nearly constant, so that defining $\nabla \lambda_i$ using (16) provides values that are close to $\mathbf{0}$. We therefore propose the following modification of (16) to maintain a correct direction of the normals:

$$\nabla \lambda_i = \begin{cases} \sum_{j \in \Omega_i} (\lambda_j - \lambda_i) (\mathbb{L}_i \nabla W_{ij}) V_j & \text{if } \lambda_i \geq \lambda_{jet} \\ \sum_{j \in \Omega_i} \lambda_j (\mathbb{L}_i \nabla W_{ij}) V_j & \text{otherwise,} \end{cases} \quad (20)$$

where λ_{jet} is a threshold value used here to detect if the particle i is surrounded by an insufficient number of neighbor particles. We set its value to $\lambda_{jet} = 0.7$. Although the correction (20) does not switch in a continuous manner, normalizing as in (15) prevents from the occurrence of discontinuities in $\mathbf{n}_i$.

3.4.2. Normal penalization far from the interface

Far from the interface the value of λ_i is close to a constant value ($\lambda_i \simeq 1$) so that $\nabla \lambda_i$ tends towards $\mathbf{0}$. As a consequence, the normals resulting from (15) will point in arbitrary directions if no correction is made in these regions. Basically for such particles, the normals must be set to $\mathbf{0}$ in order to properly compute the curvature (17). As suggested in [6], we propose here a criterion based on the norm of $\nabla \lambda_i$, leading to:

$$\mathbf{n}_i = \begin{cases} -\frac{\nabla \lambda_i}{\|\nabla \lambda_i\|} & \text{if } \|\nabla \lambda_i\| > \epsilon \frac{\lambda_i}{h_i} \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (21)$$

where h_i is the smoothing length of the kernel (which depends on the kernel radius and on the kernel function [21]) and ϵ is a constant set to $\epsilon = 0.1$ (see [6] for more justifications). Although this correction does not switch in a continuous manner, it switches approximately at a distance R from the interface, so that the use of the interfacial function $\delta_{\Sigma,i}$ (see Fig. 1) prevents from perceptible discontinuous variations of the surface tension force (14).

3.4.3. Curvature correction inside fluid jets

Another difficulty occurring in fluid jet regions resides in the presence of strong variations of the normal directions, even in presence of correctly evaluated normals, as illustrated in Fig. 3. In such cases, Eq. (17) should be modified to provide a better evaluation of the curvature. We propose the following correction:

$$\kappa_i = \sum_{j \in \Omega_i} s_{ij} (\mathbf{n}_j - \mathbf{n}_i) \cdot (\mathbb{L}_i \nabla W_{ij}) V_j, \quad (22)$$

where

$$s_{ij} = \begin{cases} 1 & \text{if } \mathbf{n}_i \cdot \mathbf{n}_j \geq \cos \alpha_t \text{ and } \|\mathbf{n}_i\| > 0 \text{ and } \|\mathbf{n}_j\| > 0 \\ 0 & \text{otherwise,} \end{cases} \quad (23)$$

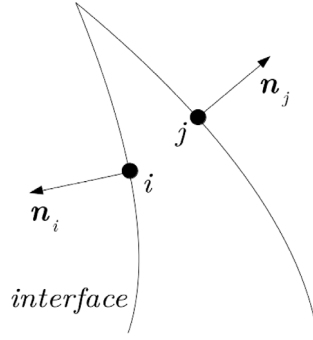


Fig. 3. Example of situation for which the correction (22) is necessary (here strong variations of the normal directions).

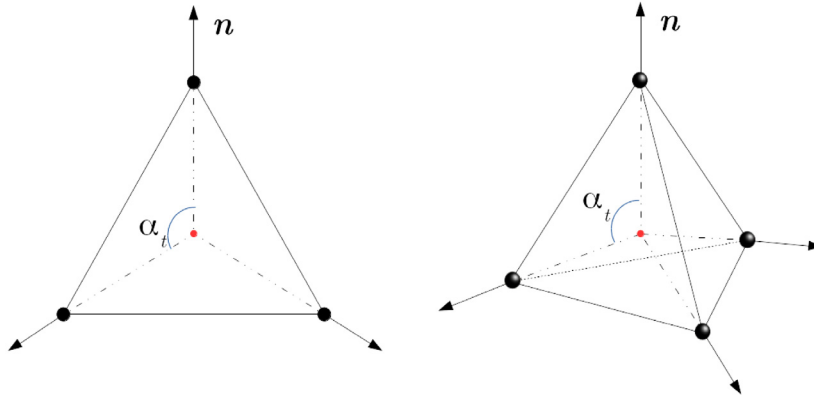


Fig. 4. Limit configurations corresponding to the smallest radius of curvature that can be detected. Left : 2D case. Right : 3D case.

where α_t is a threshold angle defined as $\cos \alpha_t = -\frac{1}{2}$ in 2D and $\cos \alpha_t = -\frac{1}{3}$ in 3D. Those threshold values correspond respectively to the angle at the center of an equilateral triangle and of a regular tetrahedron (cf. Fig. 4). This choice is made with respect to the smallest radius of curvature which can be detected (as three points are needed to fit a circle and four to fit a sphere). Although (23) does not switch in a continuous manner, no discontinuity is observed in the evaluation of curvature as it is smoothed in (22).

3.4.4. Interfacial function correction

The last proposed correction is related to the interfacial function (19). In regions where particles have a small number of neighbors (for instance at the tip of a fluid jet or near a fluid filament breakup) we observe in practice that the surface tension effects are slightly underestimated, forcing the particles to move away from the fluid domain. In order to correct this issue we propose the following modification of (19):

$$\delta_{\Sigma,i} = 2 \max(1, g_i) \left\| \sum_{j \in \Omega_i} \nabla W_{ij} V_j \right\|, \quad (24)$$

where the function g_i is greater than 1 exclusively in regions where the number of neighbors is small, such that surface tension effects could be slightly increased in these regions. We propose here the use of the Shepard correction as:

$$g_i = \frac{1}{2 \sum_{j \in \Omega_i} W_{ij} V_j}. \quad (25)$$

This function respects the desired condition as the discrete Shepard correction is close to 0.7 ($g_i \simeq 0.71$) near a flat free-surface, which does not trigger the correction (24). Conversely, this function will tend to slightly increase

the surface tension forces for lower values of the discrete Shepard correction, that is to say in fluid jet tips or fluid filament breakup for instance. Note that the Shepard correction used here should be considered as a first attempt, as any other choice respecting $g_i > 1$ in regions with small number of neighbors could be used.

3.5. Stability condition

As showed in [1], the volumic formulation (14) implies a stability condition, depending on the propagation speed of a capillary waves. For single-phase flows, the phase velocity of a capillary wave is given by:

$$c_\phi = \sqrt{\frac{\sigma k}{\rho}}, \quad (26)$$

where k is the wave number associated to the considered capillary wave.

Thus, the stability condition states that a capillary wave must cross at most a certain fraction of the spatial step size (taken here as the kernel radius R_i) during a time step Δt , leading to the following condition:

$$\Delta t \leq CFL_{ST} \min_i \left(\frac{R_i}{c_{\phi,i}} \right) = CFL_{ST} \min_i \left(\sqrt{\frac{\rho_i R_i^2}{\sigma k_i}} \right), \quad (27)$$

where CFL_{ST} is the stability constant related to the surface tension terms. The wave number k_i must now be evaluated. Brackbill et al. [1] and Morris [6] proposed to evaluate k_i with the discretization size of the method (for example [1] takes $k_i \sim \frac{2\pi}{\Delta x}$). This choice leads to a constant surface tension time step during the whole simulation, so that it does not change when strong curvatures are present or at the contrary when a more limited curvature occurs. We propose here a modification of the stability condition to account for the change of the curvature during a simulation. The wave number k_i is now evaluated using the computed curvature (17) ($k_i \sim 2\pi|\kappa_i|$), leading to:

$$\Delta t \leq CFL_{ST} \min_i \left(\sqrt{\frac{\rho_i R_i^2}{2\pi\sigma|\kappa_i|}} \right), \quad (28)$$

with a coefficient set here to $CFL_{ST} = 0.05$, optimized heuristically.

3.6. Compressibility and surface tension

The surface tension is characterized by a pressure jump at the interface. In particular, for large values of the curvature at the interface, the pressure jump could become significant, impacting the weakly-compressible hypothesis. As showed in [31], the weakly-compressible hypothesis is verified as long as the following condition is satisfied:

$$\frac{\sigma|\kappa|}{\rho c^2} < 0.01. \quad (29)$$

This condition can be obtained through an analysis of the momentum equation (4), and implies that the density variations around the reference value ρ_0 stay in the order of one percent. This condition is important to guarantee the weakly-compressible property of the solution, in particular for cases where strong curvatures occur. This is not directly imposed inside the present formulation, but must be verified a posteriori (otherwise the artificial speed of sound c_0 must be adjusted accordingly).

4. Validation for cases without solid boundaries

This section aims at validating the proposed single-phase surface tension model on various representative test cases such as pressure jump at the interface, oscillatory effects and rupture phenomenon. In all the following cases, the ratio $\frac{R}{\Delta x}$ is systematically taken as 4 for 2D cases and 3 for 3D cases, corresponding to approximately 50 and 110 neighbors per particles respectively.

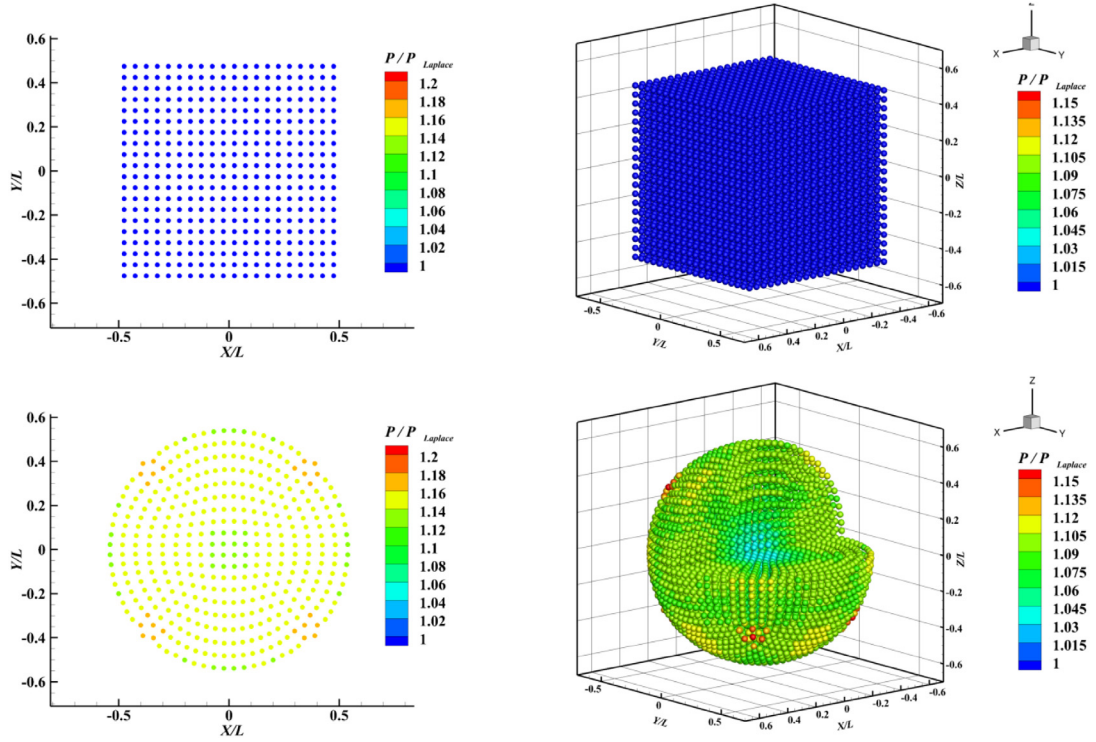


Fig. 5. Square and cubic drop deformation : particle distribution and pressure field at $t = 0$ (top) and $t = 8\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$ (bottom) for 2D case (left) and 3D case (right) obtained with the present C-CSF formulation and with $\frac{L}{\Delta x} = 20$.

4.1. Square and cubic drop deformation

The first test case consists in studying the deformation of an initial square or cubic fluid domain of size L . The chosen parameters are the same as those used in [29] : $\rho_0 = 1 \text{ kg/m}^3$, $\sigma = 1 \text{ N/m}$, $\mu = 0.2 \text{ kg/m/s}$ and $L = 0.4 \text{ m}$. The initial fluid volume V_0 domain deforms up to become a circular shape, as showed in Fig. 5. An equilibrium state is reached at $t = 8\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$ where the velocity tends to zero and the pressure to $P_{Laplace}$, theoretically defined as:

$$P_{Laplace} = \begin{cases} \frac{2\sigma}{R_{eq}} & \text{in 3D} \\ \frac{\sigma}{R_{eq}} & \text{in 2D,} \end{cases} \quad (30)$$

where R_{eq} is the radius of the drop when reaching its equilibrium state ($R_{eq} = (\frac{3}{4\pi})^{1/3} L$ in 3D and $R_{eq} = \frac{L}{\sqrt{\pi}}$ in 2D). Fig. 5 shows that the pressure field obtained inside the drop is quite uniform, with pressure values close to the expected theoretical values in (30). Peaks of error are observed at the initial corners of the fluid patches, due to a less regular particle distribution in those areas.

This case test is useful to highlight the importance of the choice of renormalized operators in the computation of the normals (15) and the curvature (17) at the interface. Fig. 6 shows the influence of the renormalized operators on the accuracy of the model. In particular Fig. 6 shows the importance of renormalized operators to avoid numerical artefacts at corner particles. The solution is also widely improved thanks to the proposed scheme corrections presented in Section 3.4, as shown in Fig. 7 where the pressure field along the x -axis is presented, showing to be more regular and close to the expected Laplace pressure (30). The maximum local error observed with the new C-CSF formulation proposed does not exceed 10%. Fig. 7 also shows that the C-CSF formulation can be successfully used in various SPH schemes such as Riemann-based SPH schemes [13] or δ -SPH schemes [14,15].

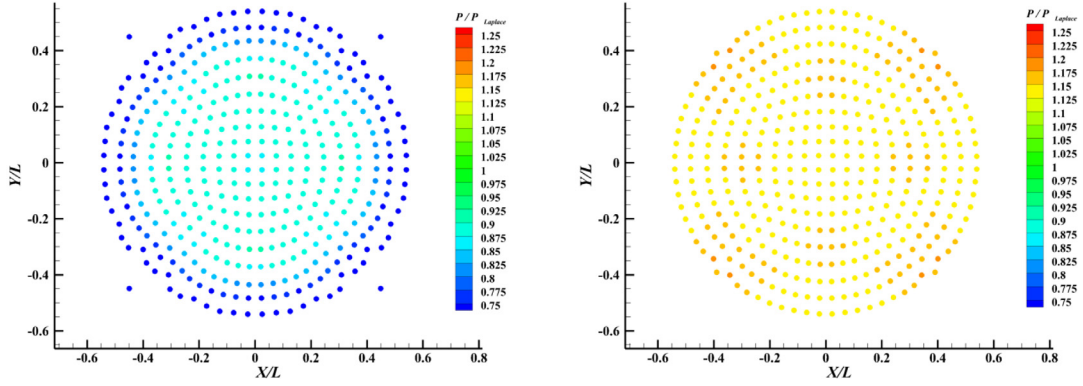


Fig. 6. Square drop deformation with $\frac{L}{\Delta x} = 20$: particle distribution and pressure field at $t = 8\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$ for non-renormalized (left) and renormalized (right) operators for the evaluation of the normals and the curvature at the interface, emphasizing the benefits from using (15) and (17).

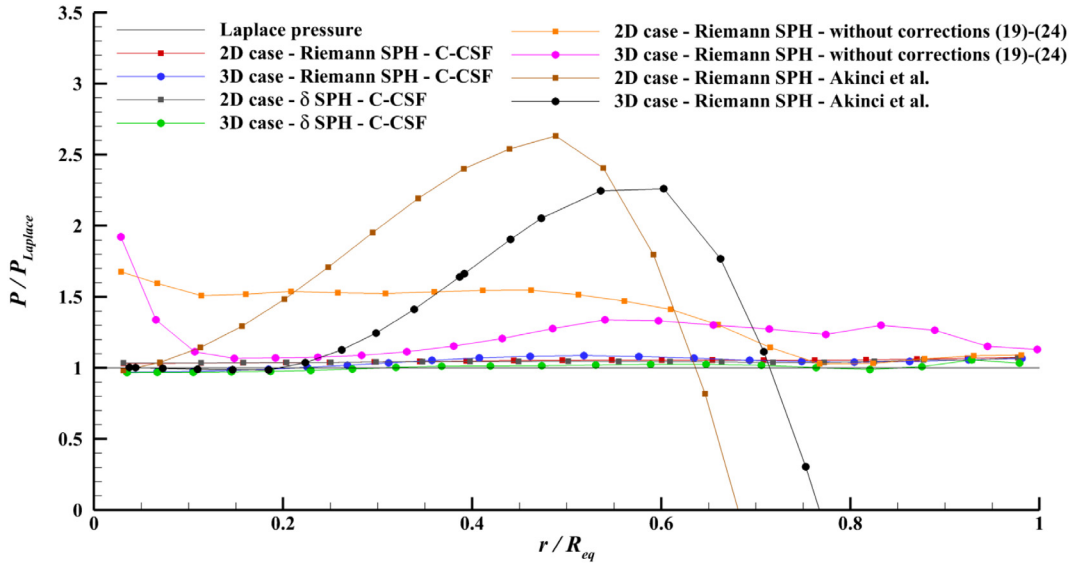


Fig. 7. Square and cubic drop deformation: pressure profiles along the x -axis for 2D and 3D cases and for $\frac{L}{\Delta x} = 40$ at $t = 8\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$. We also give the results without the corrections presented in Section 3.4, for the δ -SPH scheme and for the PF formulation from Akinci et al. [11].

Finally, Fig. 7 compares the C-CSF formulation to the Pairwise Force formulation proposed by Akinci et al. [11]. This shows in particular that the PF formulation gives wrong evaluation of the pressure field, especially near the free-surface.

The kinetic energy evolution is also presented in Fig. 8 and shows that the kinetic energy tends to a value close to zero. Note that this Figure shows that results presented above are given for an equilibrium state (reached for $t \gtrsim 2\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$). The fact that this value is not strictly zero means that some small parasitic currents occurs at the equilibrium state. Nevertheless Fig. 8 also shows that the accuracy of the model is improved as the number of particles involved increases so that the parasitic currents become negligible as the spatial resolution increases. To highlight the convergence property of the present scheme, a convergence study has been performed on the final mean pressure inside the drop, as presented in Fig. 9. Note that the results obtained without the corrections prescribed in Section 3.4 have been omitted from this Figure, as no convergence can be obtained, underlining the importance of

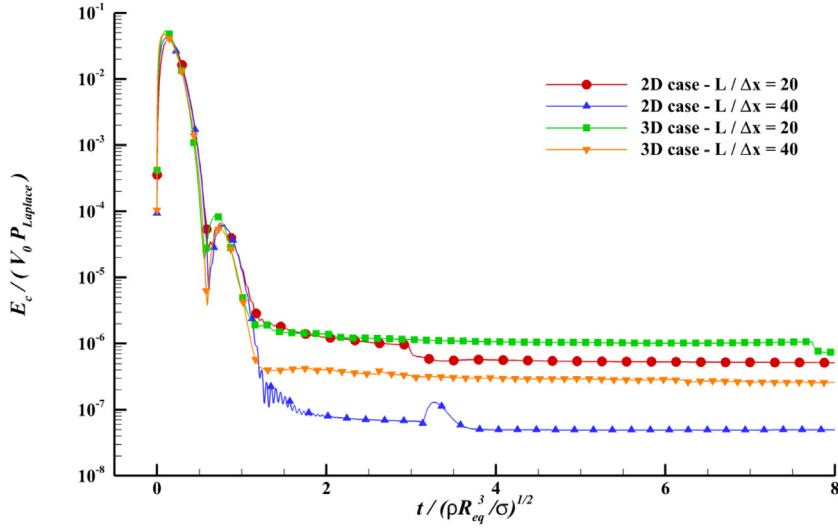


Fig. 8. Square and cubic drop deformation : kinetic energy evolution for 2D and 3D cases and for $\frac{L}{\Delta x} = 20$ and $\frac{L}{\Delta x} = 40$.

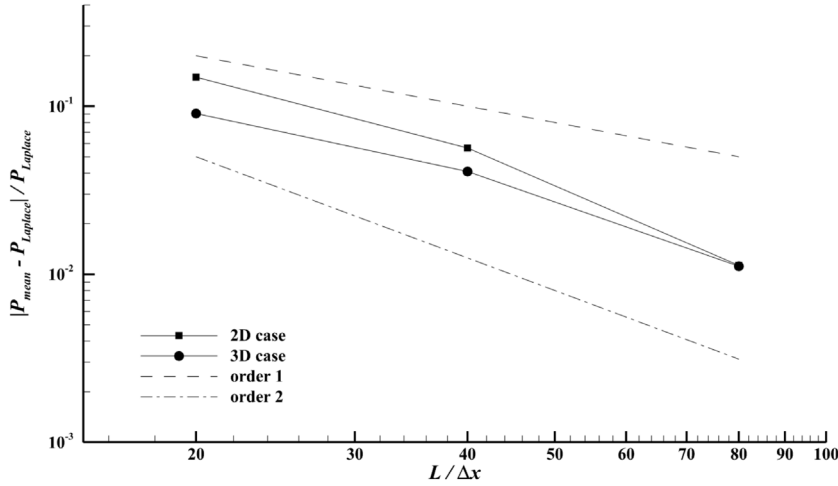


Fig. 9. Square and cubic drop deformation : Convergence curves on the mean pressure of the drop at $t = 8\sqrt{\frac{\rho R_{eq}^3}{\sigma}}$.

such corrections. The convergence rate reached is between a 1st and 2nd order for both 2D and 3D cases, showing that the convergence is maintained using the present model and that errors lower than one percent can be obtained.

4.2. Oscillating drop

The second test case is an oscillating drop under the action of surface tension. For an inviscid fluid, the analytical and linearized solution can be derived as a standing wave solution [32], based on the decomposition over the eigen modes of the drop. For the mode n , given a harmonic oscillation of the drop :

$$r_{interface} = R_0 + a H_{mn} \sin(\omega_n t + \psi), \quad (31)$$

the solution is then given both in terms of velocity potential and pressure by:

$$\phi = -\frac{a \omega_{n,\sigma} R_0}{n} H_{mn} \frac{r^n}{R_0^n} \cos(\omega_{n,\sigma} t + \psi), \quad (32)$$

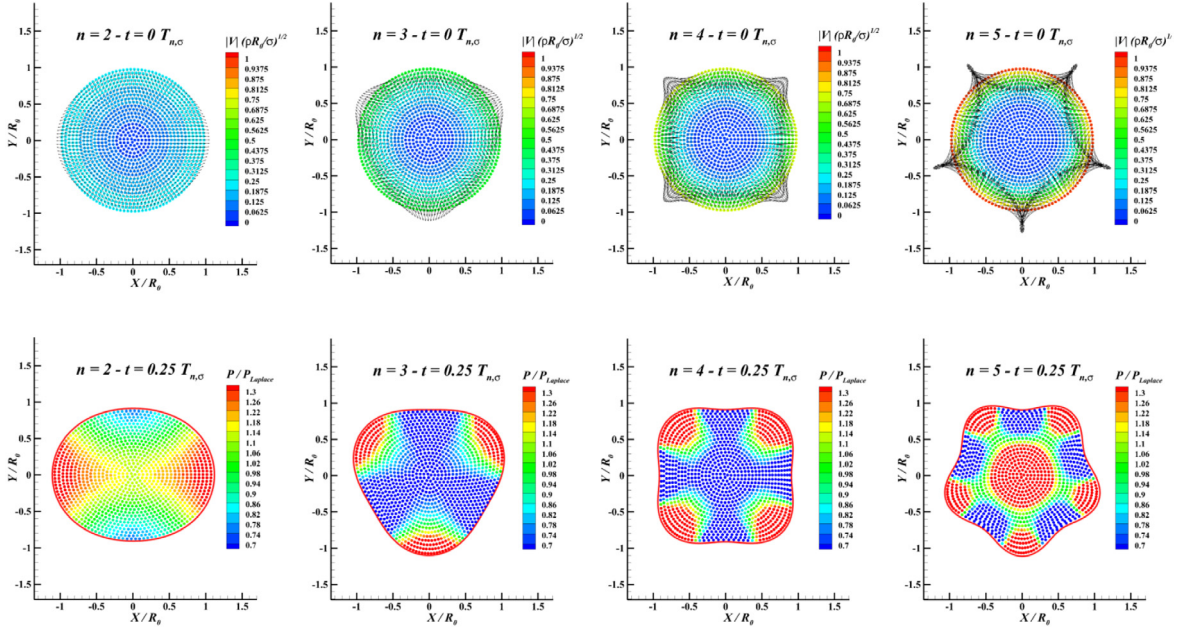


Fig. 10. Oscillating drop computed with $\frac{R_0}{\Delta x} = 20$. Top : velocity field at $t = 0$. Bottom : pressure field $t = 0.25 T_{n,\sigma}$. Red lines correspond to the theoretical shapes from Eq. (31).

$$P = \frac{a \rho \omega_{n,\sigma}^2 R_0}{n} H_{mn} \frac{r^n}{R_0^n} \sin(\omega_{n,\sigma} t + \psi) + P_{Laplace}, \quad (33)$$

where the small parameter a determines the amplitude of the oscillations, ψ an arbitrary phase, R_0 the radius of the drop at rest, r the radial coordinate ($r \in [0, r_{interface}]$) and H_{mn} is the harmonic function of order (m, n) (for 3D cases H_{mn} corresponds to the spherical harmonic Y_{mn} ; for 2D cases $H_{mn} = \cos(n\theta)$, where θ is the cylindrical coordinate). In 2D the eigen pulsation $\omega_{n,\sigma}$ is given by :

$$\omega_{n,\sigma}^2 = \frac{n(n+1)(n-1)\sigma}{\rho R_0^3}. \quad (34)$$

The following parameters are chosen : $\rho = 1 \text{ kg/m}^3$, $R_0 = 0.2 \text{ m}$, $a = 0.1 R_0$ and $\sigma = 0.5 \text{ N/m}$. Four modes are studied : $n \in [2, 3, 4, 5]$. The particle distribution and the pressure field at $t = 0.25 T_{n,\sigma}$ are presented in Fig. 10. We also compare the oscillating radius of the drop with the theoretical solution (31) (plotted as the red lines in Fig. 10). A good agreement is obtained for all first modes.

To further investigate the proposed model, the influence of the value of σ is also studied. For the first four modes, the oscillation periods are compared with the theoretical ones (obtained with (34)) in Fig. 11, showing a good agreement with the theoretical results. Computations with no corrections have also been performed but they resulted in unstable solutions, showing the necessity of the proposed corrections.

4.3. Plateau-Rayleigh instability

The Plateau-Rayleigh instability is characteristic of fluid cylinders subject to an initial perturbation. Noting $k = \frac{2\pi}{\lambda}$ the wave number associated to the perturbation and R_0 the initial radius of the fluid cylinder, it can be showed theoretically that for any perturbation such that $k R_0 \leq 1$ an instability occurs, causing the fluid cylinder to fragment into several drops [33] (see Fig. 12). In addition, it can be found that the most unstable perturbation is given by $k R_0 \simeq 0.697$ for an inviscid fluid [33]. This test case is particularly challenging because of its complex behavior. In particular, the fragmentation of the fluid domain into several drops could lead to numerical instabilities.

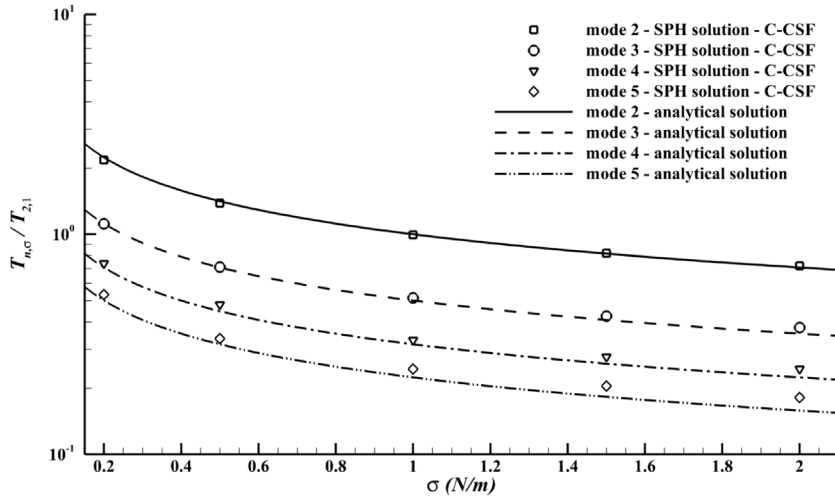


Fig. 11. Oscillating drop computed with $\frac{R_0}{\Delta x} = 20$: Evolution of the oscillation period with respect to σ for modes $n \in [2, 3, 4, 5]$.

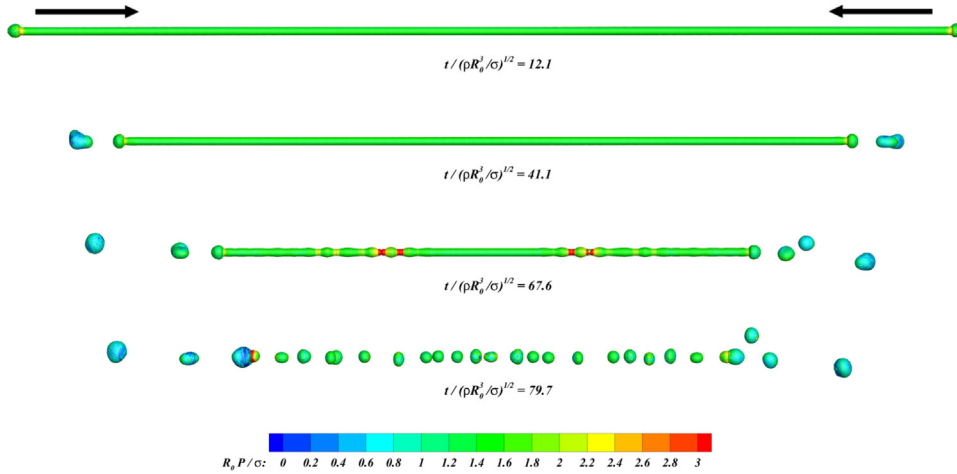


Fig. 12. Plateau-Rayleigh instability (inviscid fluid): Propagation of initial perturbations at the extremities of the fluid cylinder leading to the fragmentation into several drops, for $\frac{R_0}{\Delta x} = 4$.

This test case is initialized as a long fluid cylinder of length L . This initial shape has sharp edges at its extremities, so that perturbations are created and propagated from these extremities.

The chosen physical parameters are : $R_0 = 0.005$ m, $L = 2$ m, $\rho = 1000$ kg/m³ and $\sigma = 0.0728$ N/m and the fluid is assumed as inviscid. Results of the simulation are showed in Fig. 12, showing that the fragmentation can be obtained while avoiding numerical instabilities.

To further analyze the behavior of this case we determine the value of the most unstable perturbation. To do that we perform a Fourier transform of the free-surface deformation (determined as the radial coordinate of the cylinder interface) just before the fragmentation. Results are showed in Fig. 13 for three different spatial resolutions. The Fourier transform shows a maximal value, characteristic of the most unstable perturbation. This peak occurs at $k R_0 \simeq 0.687$ for $\frac{R_0}{\Delta x} = 4$ which is quite close to the expected theoretical value $k R_0 \simeq 0.697$.

The above study involving an inviscid fluid can be extended to the viscous case [33], showing in particular that the wave number of the most unstable perturbation reduces as the viscosity increases. We test here the influence of the viscosity on the proposed surface tension model. A similar analysis using Fourier transform has been made for cases with a viscous fluid. Fig. 14 shows the non-dimensional wave number of the most unstable perturbation as a

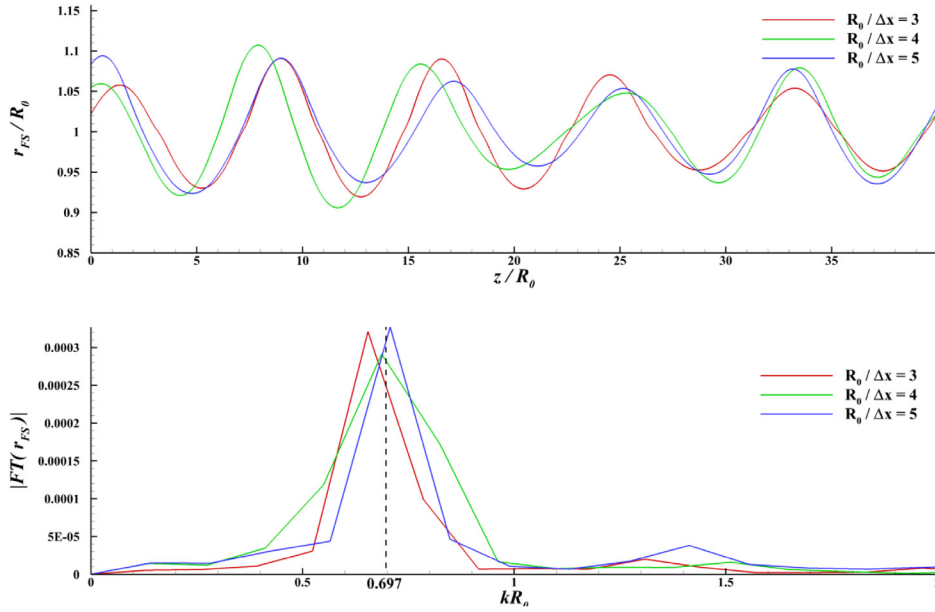


Fig. 13. Plateau-Rayleigh instability (inviscid fluid): Zoom on the free-surface deformation just before fragmentation (top) and corresponding Fourier transform (bottom) for three spatial resolutions.

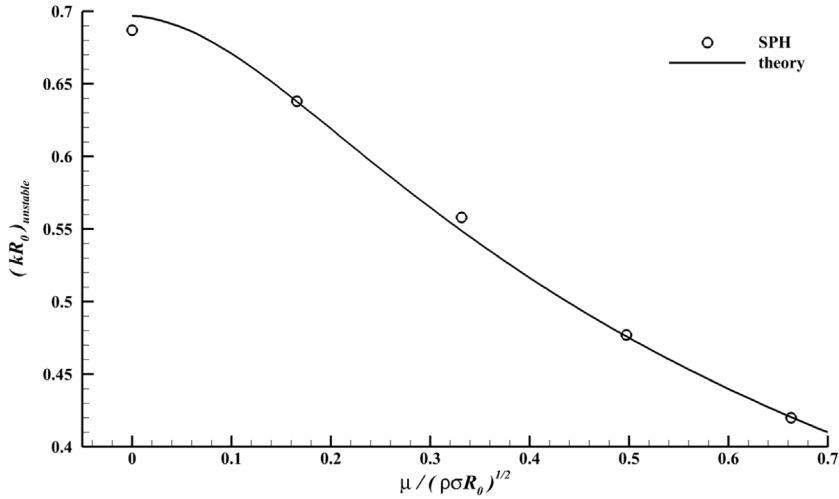


Fig. 14. Plateau-Rayleigh instability : non-dimensional wave number of the most unstable perturbation versus the non-dimensional viscosity.

function of the non-dimensional viscosity. A good agreement with theoretical results is obtained, showing that the addition of the viscous terms does not degrade the present surface tension model.

5. Taking into account solid boundaries

We now extend the C-CSF formulation to take into account the presence of solid boundaries. To this end, two formalisms are classically used in the SPH method : the ghost particle technique [14,34,35] or the use of surfacic terms in the SPH operators (such as the formalism presented in [16] and referred here to Boundary Integral Method (BIM)). The first one is often used by SPH practitioners but could suffer from limitations for problems involving complex geometries for which the BIM formalism can be preferred [16]. Therefore in the present work we preferably address this second technique although the ghost particle technique is also discussed partly, showing that the C-CSF

formulation can be used for a generic boundary condition formulation. A special attention is paid to imposing the contact angle at the contact line. In this paper, only the equilibrium contact angle is studied, dynamic contact angles being postponed to future investigations.

5.1. Ghost particle formalism

The model discussed in the previous sections remains theoretically valid with the ghost particle formalism, by summing the interactions in SPH operators not only over fluid particles but also over ghost ones. By mirroring the ghost particles symmetrically to the solid boundary, an equilibrium contact angle of $\theta_e = 90^\circ$ is intrinsically achieved. Therefore, only cases involving an equilibrium contact angle of $\theta_e = 90^\circ$ are considered while using ghost particles in the next sections.

5.2. Boundary Integral Model (BIM)

The boundary condition technique addressed here is the Cutface approach (BIM-CFA) detailed in Chiron et al. [16], which belongs to the Boundary Integral models (BIM) [17,18]. The BIM formalism is summarized hereafter (we refer the reader to the paper from Chiron et al. [16] for complementary details), and consists in computing the surfacic terms in SPH operators while applying the required boundary conditions. In addition of the advantages provided by this formulation in terms of accuracy [16], it also allows one to impose any desired contact angle, contrary to the ghost formalism. It leads to the following modifications of the numerical scheme (1)–(4):

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{u}_i + \delta\mathbf{u}_i, \quad (35)$$

$$\frac{dV_i}{dt} = \frac{V_i}{\gamma_i} \sum_{j \in \Omega_i} (\mathbf{u}_j - \mathbf{u}_i) \cdot \nabla W_{ij} V_j + \frac{V_i}{\gamma_i} \sum_{j \in \partial\Omega_i} (\mathbf{u}_j - \mathbf{u}_i) \cdot \mathbf{n}_{w,j} W_{ij} S_j + \Pi_i^{(V)}, \quad (36)$$

$$\frac{d(V_i \rho_i)}{dt} = 0, \quad (37)$$

$$\begin{aligned} \frac{d(V_i \rho_i \mathbf{u}_i)}{dt} = & -\frac{V_i}{\gamma_i} \sum_{j \in \Omega_i} (P_j + P_i) \nabla W_{ij} V_j - \frac{V_i}{\gamma_i} \sum_{j \in \partial\Omega_i} (P_j + P_i) W_{ij} \mathbf{n}_{w,j} S_j \\ & + V_i \rho_i \mathbf{g} + V_i \mathbf{F}_i^\mu + V_i \mathbf{F}_i^\sigma + \Pi_i^{(\rho V \mathbf{u})}, \end{aligned} \quad (38)$$

where S_j is the area of the discretized boundary face j , $\mathbf{n}_{w,j}$ is the normal to the face j (pointing from the fluid to the boundary). The expression of the stabilization terms $\Pi_i^{(V)}$ and $\Pi_i^{(\rho V \mathbf{u})}$ depends on the numerical scheme [19]. As detailed in [16], each face j does not necessarily correspond to the face of the input mesh. In practice, each face of the input mesh is divided into several smaller faces j . The velocity $\mathbf{u}_j$ at the face j in (36) depends on whether a no-slip or a free-slip boundary condition is applied ($\mathbf{u}_j = \mathbf{u}_w$ in case of a no-slip condition and $\mathbf{u}_j \cdot \mathbf{n}_{w,j} = \mathbf{u}_w \cdot \mathbf{n}_{w,j}$ in case of a free-slip condition, where $\mathbf{u}_w$ is the boundary velocity). The pressure P_j at the face j in (38) is given by $P_j = P_i + \rho_i c_i (\mathbf{u}_w - \mathbf{u}_i) \cdot \mathbf{n}_{w,j}$ [16]. γ_i corresponds to the Shepard correction but is computed analytically ; it mainly depends on the distance to the boundary [16].

In such a formalism an expression of the Laplacian operator from Morris et al. [27] has been proposed in [16]. It represents the viscous force $\mathbf{F}_i^\mu$ and is expressed as:

$$\Delta^{MEA} \mathbf{u}_i = \frac{2}{\gamma_i} \sum_{j \in \Omega_i} \frac{(\mathbf{u}_i - \mathbf{u}_j)}{\|\mathbf{x}_i - \mathbf{x}_j\|^2} (\mathbf{x}_i - \mathbf{x}_j) \cdot \nabla W_{ij} V_j + \frac{2}{\gamma_i} \sum_{j \in \partial\Omega_i} \frac{(\mathbf{u}_i - \mathbf{u}_w)}{d_{n,ij}} W_{ij} S_j, \quad (39)$$

where $d_{n,ij}$ is the normal distance between particle i and face j . The expression of $\mathbf{F}_i^\sigma$ is the same as in (14) and its adaptation to the BIM formalism is detailed below.

5.2.1. Renormalized operators in the BIM formalism

As showed for cases without boundaries, renormalized operators provide a better accuracy in the context of single-phase SPH. In BIM formalism, Chiron already proposed a method to renormalize the SPH operators [36]:

$$\nabla f_i = \mathbb{L}_i \left(\frac{1}{\gamma_i} \sum_{j \in \Omega_i} (f_j - f_i) \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial\Omega_i} (f_j - f_i) \mathbf{n}_{w,j} W_{ij} S_j \right), \quad (40)$$

where $\mathbb{L}_i$ is computed such that the gradient of a linear field is constant (i.e. such that the first order consistency is obtained), leading to:

$$\mathbb{L}_i = \left(\frac{1}{\gamma_i} \sum_{j \in \Omega_i} (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial \Omega_i} (\mathbf{x}_j - \mathbf{x}_i) \otimes \mathbf{n}_{w,j} W_{ij} S_j \right)^{-1}. \quad (41)$$

Notice that without boundaries, expressions (9) and (40) are equivalent (assuming $\gamma_i = 1$). For a vector field, (40) becomes:

$$\nabla \mathbf{f}_i = \mathbb{L}_i \left(\frac{1}{\gamma_i} \sum_{j \in \Omega_i} (\mathbf{f}_j - \mathbf{f}_i) \otimes \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial \Omega_i} (\mathbf{f}_j - \mathbf{f}_i) \otimes \mathbf{n}_{w,j} W_{ij} S_j \right). \quad (42)$$

The divergence operator is then defined as the trace of the gradient operator (42). Then, it can be showed that the renormalized divergence operator can be written as:

$$\nabla \cdot \mathbf{f}_i = \frac{1}{\gamma_i} \sum_{j \in \Omega_i} (\mathbb{L}_i (\mathbf{f}_j - \mathbf{f}_i)) \cdot \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial \Omega_i} (\mathbb{L}_i (\mathbf{f}_j - \mathbf{f}_i)) \cdot \mathbf{n}_{w,j} W_{ij} S_j. \quad (43)$$

5.2.2. Single-phase surface tension model in the BIM formalism

To compute all terms of the surface tension force (14) in the BIM formalism, first the normal to the interface $\mathbf{n}_i$ is still computed using (15) but with the minimum eigenvalue of the matrix $\mathbb{L}_i^{-1}$ defined in (41), and where:

$$\mathbf{n}_i = -\frac{\nabla \lambda_i}{\|\nabla \lambda_i\|}, \quad (44)$$

and where

$$\nabla \lambda_i = \mathbb{L}_i \left(\frac{1}{\gamma_i} \sum_{j \in \Omega_i} (\lambda_j - \lambda_i) \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial \Omega_i} (\lambda_j - \lambda_i) \mathbf{n}_{w,j} W_{ij} S_j \right). \quad (45)$$

Note that Eq. (45) requires the values λ_j for $j \in \partial \Omega_i$. In a first attempt, and for the sake of simplicity, these values are set to $\lambda_j = 1$ in the present work, as proposed by Sun et al. [37] for the “fixed ghost particles” formalism.

The curvature κ_i at the interface is still computed using the renormalized divergence of the normal $\mathbf{n}_i$ to the interface, leading here to:

$$\kappa_i = \frac{1}{\gamma_i} \sum_{j \in \Omega_i} (\mathbb{L}_i (\mathbf{n}_j - \mathbf{n}_i)) \cdot \nabla W_{ij} V_j + \frac{1}{\gamma_i} \sum_{j \in \partial \Omega_i} (\mathbb{L}_i (\mathbf{n}_j - \mathbf{n}_i)) \cdot \mathbf{n}_{w,j} W_{ij} S_j, \quad (46)$$

where the scalar product $\mathbf{n}_j \cdot \mathbf{n}_{w,j}$ can be used to impose the contact angle at the contact line. For a constant equilibrium contact angle θ_e , we set $\mathbf{n}_j \cdot \mathbf{n}_{w,j} = -\cos \theta_e$.

Moreover, the color gradient is computed using:

$$\nabla c_i = \frac{2}{\gamma_i} \sum_{j \in \Omega_i} \nabla W_{ij} V_j + \frac{2}{\gamma_i} \sum_{j \in \partial \Omega_i} \mathbf{n}_{w,j} W_{ij} S_j. \quad (47)$$

5.3. Additional corrections

As for the case without solid boundaries, some corrections are needed to make the model more accurate and more robust. First, all the corrections detailed in Section 3.4 still hold in the BIM formalism. Then, the approximation made by setting $\lambda_j = 1$ at the boundary leads to relatively erroneous estimated normals at the interface near the contact line. So, it is necessary to correct the evaluation of the normal $\mathbf{n}_i$ near the boundary. Although it is relatively erroneous at first glance, we observed that nevertheless $\mathbf{n}_i$ provides a good estimation of the tangent vector $\mathbf{t}_{b,i}$ to the boundary (and normal to the contact line) :

$$\mathbf{t}_{b,i} = \frac{\mathbf{n}_i - (\mathbf{n}_i \cdot \mathbf{n}_{b,i}) \mathbf{n}_{b,i}}{\|\mathbf{n}_i - (\mathbf{n}_i \cdot \mathbf{n}_{b,i}) \mathbf{n}_{b,i}\|}. \quad (48)$$

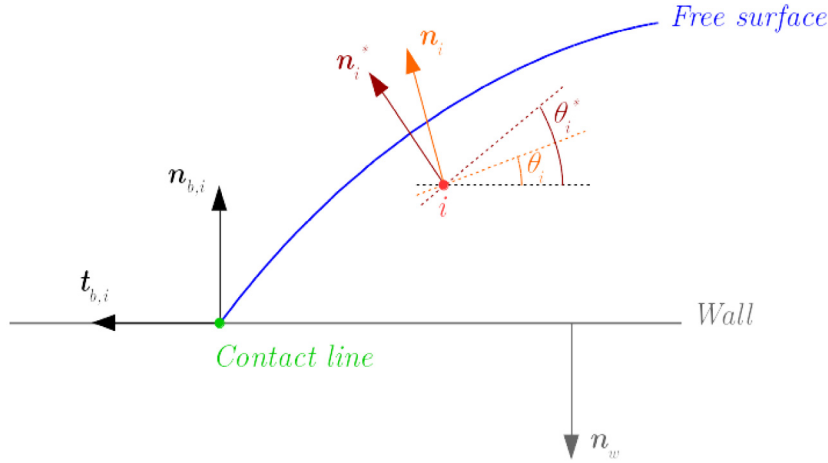


Fig. 15. Parametrization used for the correction of the normal near the contact line (here represented in the plane $(t_{b,i}, n_{b,i})$).

where $n_{b,i}$ is the normal vector of the closest face to the particle i and pointing from the boundary to the fluid (cf. Fig. 15). This tangent vector can then be used to correct the evaluation of the normal while imposing a correct contact angle. Inspired by the correction proposed by Breinlinger et al. [8], we apply the following smoothed correction near to a boundary :

$$n_i^* = \cos\theta_i^* n_{b,i} + \sin\theta_i^* t_{b,i}, \quad (49)$$

with θ_i^* the corrected contact angle.

Finally, we apply the following quadratic correction (chosen empirically) to the contact angle:

$$\theta_i^* = \begin{cases} \theta_e + (\theta_i - \theta_e) \left(\frac{d_i}{R_i} \right)^2 & \text{if } d_i < R_i \\ \theta_i & \text{otherwise,} \end{cases} \quad (50)$$

where R_i is the kernel radius of the particle i , d_i the distance to the nearest boundary and θ_i the contact angle before correction and given by $\theta_i = \arccos(n_i \cdot n_{b,i})$. This correction is made before the computation of the curvature (46) and n_i is then substituted by n_i^* in (14) and (46).

6. Validation of cases involving solid boundaries

Two cases are considered here for validating the proposed single-phase surface tension model in presence of solid boundaries. Notice here that a tangential particle shifting at the free-surface is used to improve the simulations, especially near the contact line (see [15] for the details of this special treatment at the free-surface). Here again, a particular attention is paid to the accuracy of the proposed model.

6.1. Static drop on a flat surface

The first case consists in capturing the shape of a static drop on a flat surface. Its shape theoretically depends on: the surface tension coefficient, the volume of the drop, the gravity acceleration and the equilibrium contact angle. We propose here to validate our present model over different contact angle values. In particular we compare the measured height of the drops with the theoretical ones, given by solving the following differential equation [38]:

$$-\frac{\bar{r}''}{(1 + \bar{r}'^2)^{3/2}} + \frac{1}{\bar{r}(1 + \bar{r}'^2)^{1/2}} = 2 + \left(\frac{b}{R_0} \right)^2 Bo \bar{z}, \quad (51)$$

where $Bo = \frac{R_0^2 \rho g}{\sigma}$ is the Bond number of the drop, R_0 the radius of the drop seen ideally as spherical (i.e. $V_0 = \frac{4}{3}\pi R_0^3$ where V_0 is the volume of the drop) and b a parameter which ensures the volume conservation of the drop (this

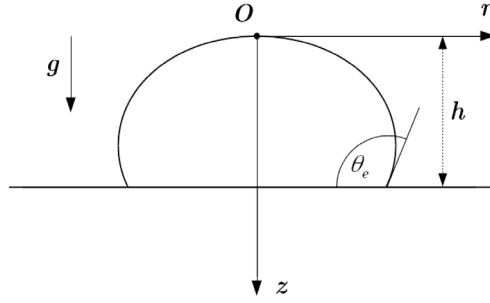


Fig. 16. Static drop on a flat surface : parametrization of the problem.

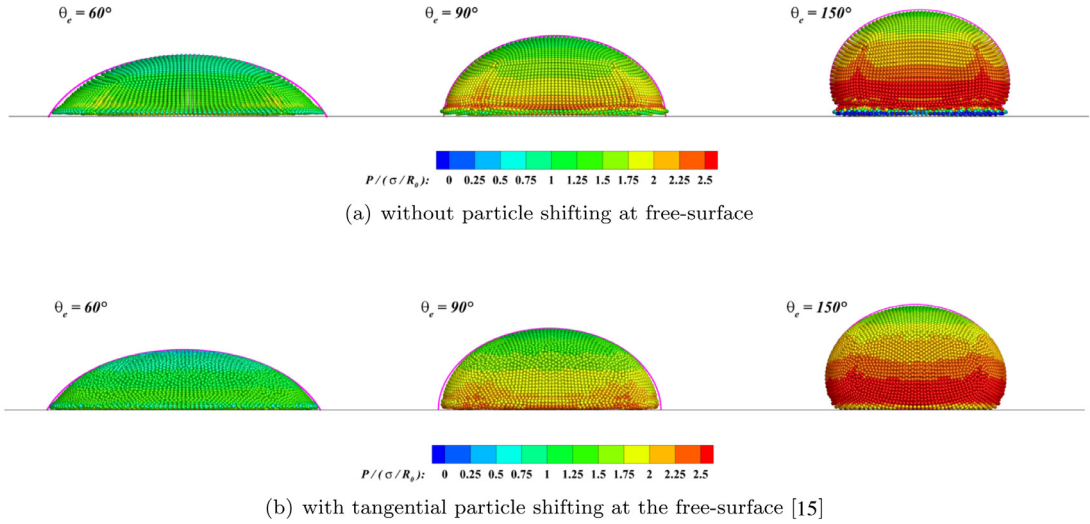


Fig. 17. Static drop on a flat surface : shape of the drop for $Bo = 1.11$, $\frac{L}{\Delta x} = 40$ and for different contact angles θ_e at $t = 8.45\sqrt{\frac{\rho R_0^3}{\sigma}}$ and without 17(a) and with 17(b) particle shifting at free-surface; magenta lines correspond to the theoretical shapes of the drops provided by (51).

parameter is not known a priori and must be determined through an iterative resolution of (51)). $(\bar{r} = \frac{r}{b}, \bar{z} = \frac{z}{b})$ is a non-dimensional parametrization of the problem (see Fig. 16 for the position of the axis and the origin). This equation is solved numerically with the following initial condition : $\bar{r}(\bar{z} = 0) = 0$ and $\bar{r}'(\bar{z} = 0) = +\infty$ (in practice the derivative at the origin is set to a sufficiently large value). The equilibrium contact angle θ_e is taken into account by taking $\bar{r}' = \cot \theta_e$ at the contact line.

The numerical setup consists in initializing a prismatic fluid patch of size $L \times L \times \frac{L}{2}$ that is let free to evolve up to an equilibrium state. The following parameters are chosen : $\rho = 1000 \text{ kg/m}^3$, $\sigma = 0.07 \text{ N/m}$, $L = 0.008 \text{ m}$, $\theta_e \in [60, 90, 150]^\circ$ and $g \in [1, 5, 10, 15] \text{ m/s}^2$ (leading respectively to $Bo \in [0.22, 1.11, 2.22, 3.32]$).

Fig. 17 shows the shape of the drops obtained, for $Bo = 1.11$ and various contact angles. We also show in this Figure that tangential particle shifting at the free-surface is necessary to improve the accuracy of the simulations, especially near the contact line. The drop height is also plotted in Fig. 18 for different contact angles and Bond numbers. A good agreement with the theoretical solution (51) is obtained: the deviations from the theoretical heights are around 5%, despite some differences near the contact lines. Those differences could be explained by the approximation made for λ_j (45) at the boundary.

The kinetic energy evolution tends to a non-zero value, due to the presence of parasitic currents and responsible for the errors made near the contact line. The values of the kinetic energy reached at the equilibrium state are given in Tables 1–3, for various Bond numbers, spatial resolutions and equilibrium contact angles (we give here only partial results but the trends are analogous with other parameters). These results show that the kinetic energy at

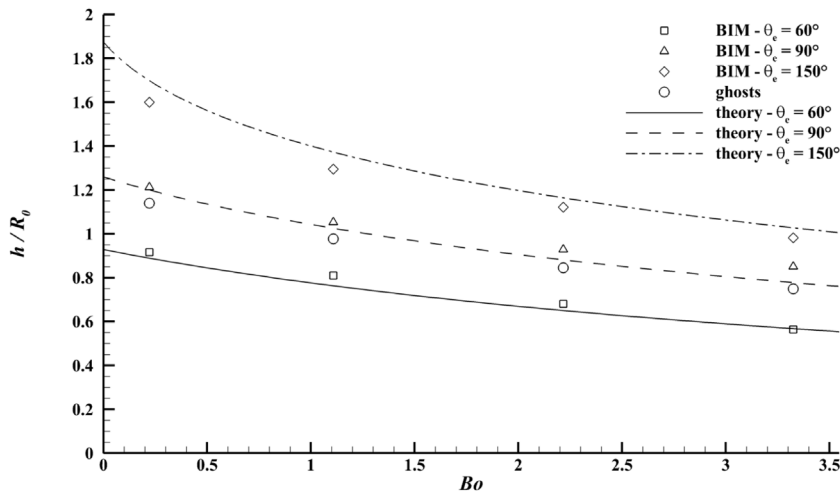


Fig. 18. Static drop on a flat surface: drop height with respect to the Bond number for $\frac{L}{\Delta x} = 20$.

Table 1

Static drop on a flat surface: non-dimensional kinetic energy at $t = 30\sqrt{\frac{\rho R_0^3}{\sigma}}$ versus Bond number, for $\frac{L}{\Delta x} = 20$ and $\theta_e = 60^\circ$.

Bo	$E_c / (\sigma V_0 / R_0)$
0.22	$2.15 \cdot 10^{-6}$
1.11	$4.26 \cdot 10^{-6}$
2.22	$6.51 \cdot 10^{-6}$
3.32	$7.89 \cdot 10^{-6}$

Table 2

Static drop on a flat surface: non-dimensional kinetic energy at $t = 30\sqrt{\frac{\rho R_0^3}{\sigma}}$ versus spatial discretization, for $Bo = 1.11$ and $\theta_e = 60^\circ$.

$L/\Delta x$	$E_c / (\sigma V_0 / R_0)$
10	$6.57 \cdot 10^{-6}$
20	$4.26 \cdot 10^{-6}$
40	$2.09 \cdot 10^{-6}$

Table 3

Static drop on a flat surface: non-dimensional kinetic energy at $t = 30\sqrt{\frac{\rho R_0^3}{\sigma}}$ versus equilibrium contact angle, for $Bo = 1.11$ and $\frac{L}{\Delta x} = 20$.

θ_e ($^\circ$)	$E_c / (\sigma V_0 / R_0)$
60	$4.26 \cdot 10^{-6}$
90	$7.06 \cdot 10^{-6}$
150	$5.72 \cdot 10^{-5}$

the equilibrium state increases as the Bond number, the particle size and the equilibrium contact angle increase. In particular, Table 2 shows that the parasitic currents tend to decrease as the number of particles increases.

The convergence rate of the present model is also highlighted in Fig. 19, showing the error made on the drop height with respect to the spatial resolution, for two different Bond numbers. The rate of convergence obtained lies between 1st and 2nd order, proving the convergence of the proposed model in presence of solid boundaries.

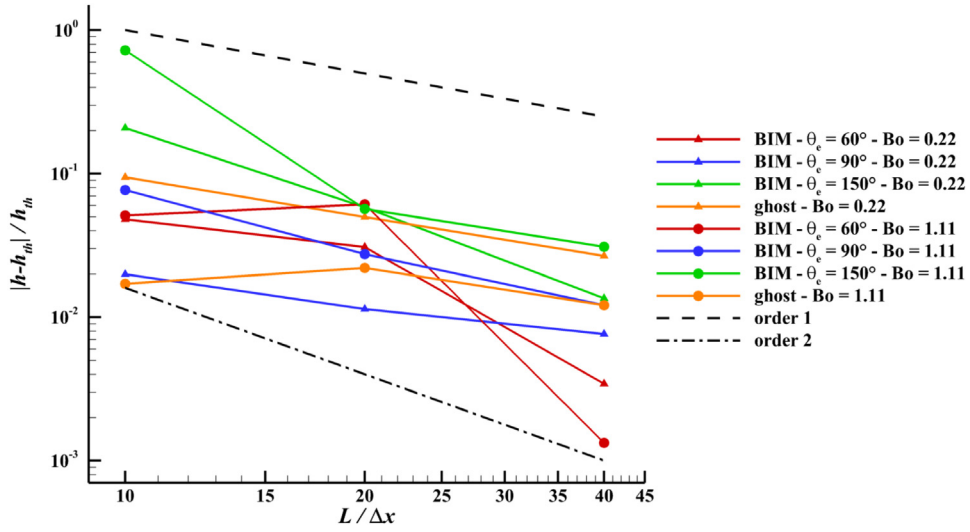


Fig. 19. Static drop on a flat surface : convergence curves related to the drop height for $Bo = 0.22$ and $Bo = 1.11$ at $t = 8.45\sqrt{\frac{\rho R_0^3}{\sigma}}$.

6.2. Drop impacting a flat surface

We now aim at validating the present surface tension model on a more complex case, involving the dynamic evolution of a spherical drop impacting a flat surface. Note that the dynamic contact angle is not treated for now in this model. Experimental and numerical results are available, as presented by Sikalo et al. [39]. The fluid characteristics are $\rho = 996 \text{ kg/m}^3$, $\sigma = 0.073 \text{ N/m}$, $\mu = 0.001 \text{ kg/m.s}$. The impact velocity is set as $U = 1.64 \text{ m/s}$ and the initial drop diameter is set to $D = 2.454 \text{ mm}$. Those parameters lead to the following Reynolds and Weber numbers : $Re = 4010$ et $We = 90$. The flat surface is assumed to be made of wax, implying an equilibrium contact angle $\theta_e \simeq 100^\circ$ [39]. Note that the equilibrium contact angle is relatively close to 90° in this test case, so that it can also be modeled using ghost particles.

This case is rather challenging due to the relatively high impact velocity imposed initially to the drop. This high impact velocity forces the drop to spread up before to become a thin film of fluid. After reaching a maximal expansion size, the drop retracts as a more compact shape (see [39] to have a full description of the test case). The results of this simulation are presented in Figs. 20 and 21 for $\frac{D}{\Delta x} = 50$. A noteworthy phenomenon occurs when the maximum spreading diameter is nearly reached: a loss of rotational symmetry characterized by circumferential instabilities at the rim is observed, which is predicted by the experiment [39], showing the capability of our model to capture complex phenomena.

Sikalo et al. [39] provide reference results on the measurement of the contact area diameter of the drop. The measurements obtained with SPH simulation (both in ghost and BIM formalisms) are showed in Fig. 22. A good agreement is obtained, both with BIM and ghost formulations of the boundary conditions (despite the ghost formulation is not suitable for cases with contact angles far from 90°). In Fig. 20 we observed some small discrepancies near the contact line which could be explained by the approximation made for $\lambda_j(45)$ at the boundary. This approximation can also explain the differences observed in Fig. 22. Another explanation to these differences can be the convergence of the simulation.

7. Conclusion

In this work, an accurate surface tension model for single-phase SPH simulations called C-CSF has been presented, available for various SPH schemes (such as Riemann SPH or δ -SPH ; although the validation with an incompressible SPH formulation has not been made in this work, we expect that this C-CSF will be available for such numerical schemes). Based on the CSF formulation [1], we showed that a particular attention must be paid to the accuracy of the SPH operators, and especially that the renormalization of the SPH operators has a critical

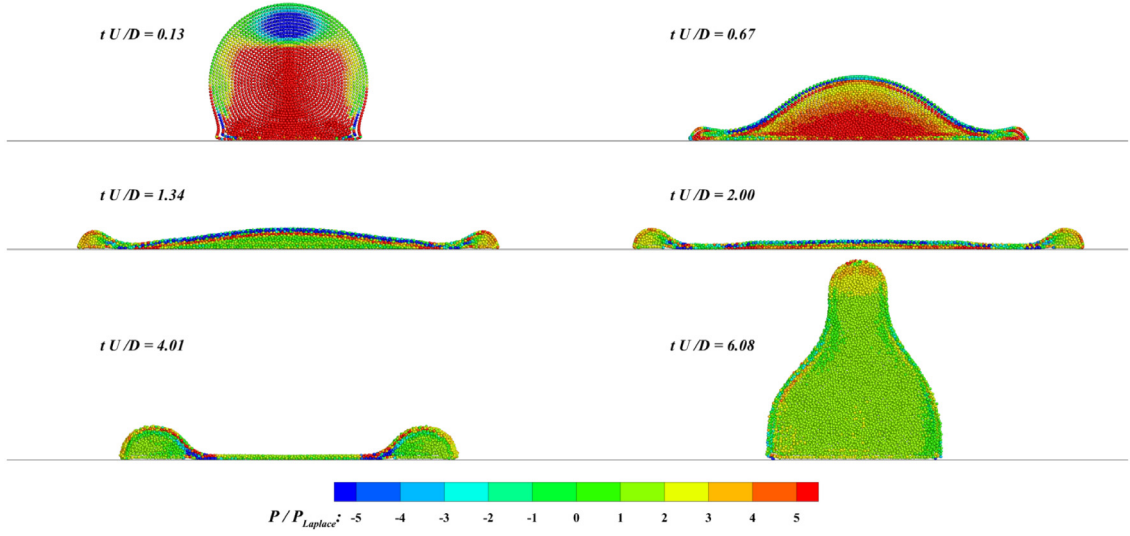


Fig. 20. Drop impacting a flat surface $\frac{D}{\Delta x} = 50$: slice view.

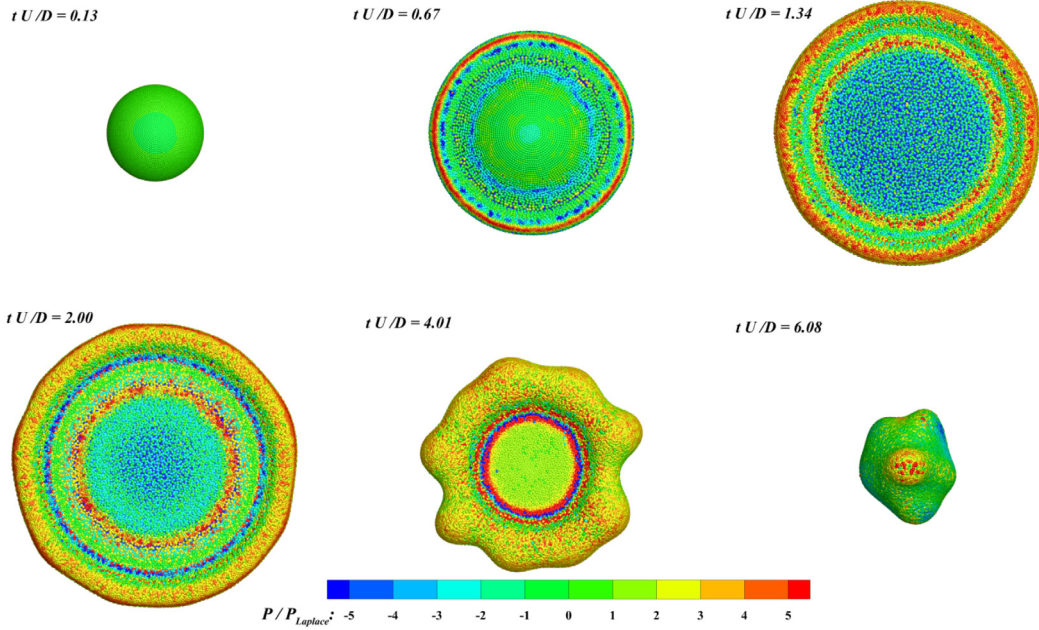


Fig. 21. Drop impacting a flat surface $\frac{D}{\Delta x} = 50$: top view.

influence. Although being more costly in terms of computational time, this approach is necessary to obtain more stable and robust simulations while increasing the accuracy of the model. In addition, we showed that some critical corrections are necessary to have more accurate results, and we expected that these corrections could be used to improve multiphase flows simulations in SPH. An extension of the model to solid boundaries problems has been proposed using a surface integration method (BIM). In particular, a simple approach has been proposed to impose an equilibrium contact angle near the contact line. The model has been validated using various test cases involving a large range of physical phenomena (static, dynamic cases). Yet some lacks of precision have been outlined, especially near the contact line. This may result in the occurrence of parasitic currents, probably due to the approximations made to impose the contact angle. Future works will be devoted to improve the proposed model.

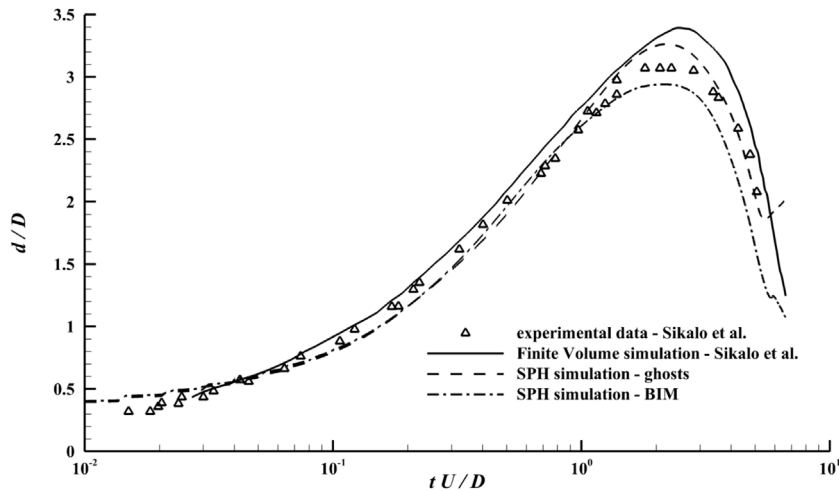


Fig. 22. Drop impacting a flat surface with $\frac{D}{\Delta x} = 50$: evolution of the contact area diameter.

in order to reduce these parasitic currents near the contact line while not only imposing equilibrium contact angles but also dynamic contact angles.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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